Homomorphic Signatures for Polynomial Functions

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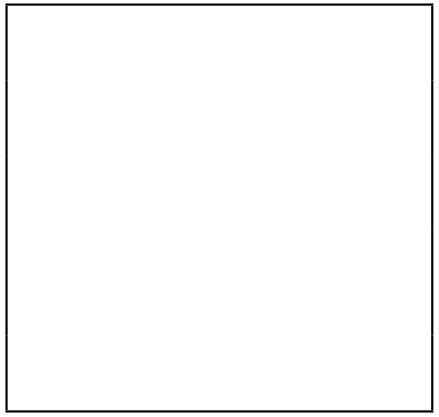
Untrusted DB



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$\xrightarrow[\text{grades}]{\text{encrypted}}$

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→ c

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- Validity: c decrypts to the correct mean.
- Security: no adversary can obtain any info about scores.
- Length efficiency: c is short.
- Privacy: decrypted mean reveals nothing else about data.

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$\sigma_1 =$ signature on
("grades", 91, "student 1")

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$\sigma =$ signature on (“grades”, 87.3, “mean”)

- Validity: σ authenticates 87.3 as the correct mean.
- Security: no adversary can authenticate a different mean.
- Length efficiency: σ is short.
- Privacy: signature reveals nothing else about data.

More generally: $\mathcal{F}$-homomorphic signatures

- $\mathcal{F}$ is a set of "admissible" functions on messages.
- τ is a "tag" tying together data from the same set.
  (like a filename)
- Given pk, admissible function $f \in \mathcal{F}$, and signatures on data

$$m_1, \ldots, m_k,$$

anyone can compute a valid signature on

$$(\tau, f(m_1, \ldots, m_k), \langle f \rangle),$$

where $\langle f \rangle$ "encodes" the function f.

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What are $\mathcal{F}$-homomorphic signatures good for?

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Arbitrary circuits	Non-linear estimators and regression Data mining (decision trees, SVM, etc.)

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Specifically, we construct secure, length-efficient, $\mathcal{F}$-homomorphic signatures for

$$\mathcal{F} = \{\text{polynomials of bounded degree with small coefficients}\}$$

Outline

1. Homomorphic signatures:
   What are they and what are they good for?

2. Linearly homomorphic signatures.

3. Polynomially homomorphic signatures from ideal lattices.

Linearly Homomorphic Signatures

Building block: GPV Signatures

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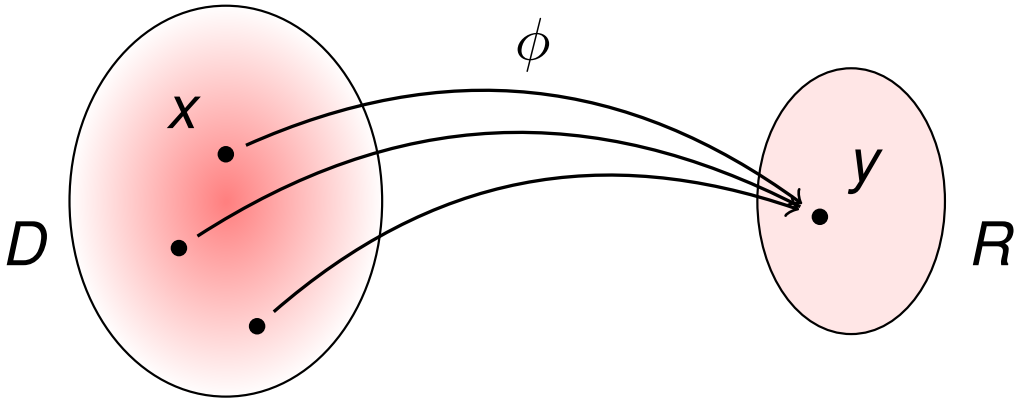
- Public function $\phi\colon D \to R$ with secret "trapdoor" ϕ^{-1}



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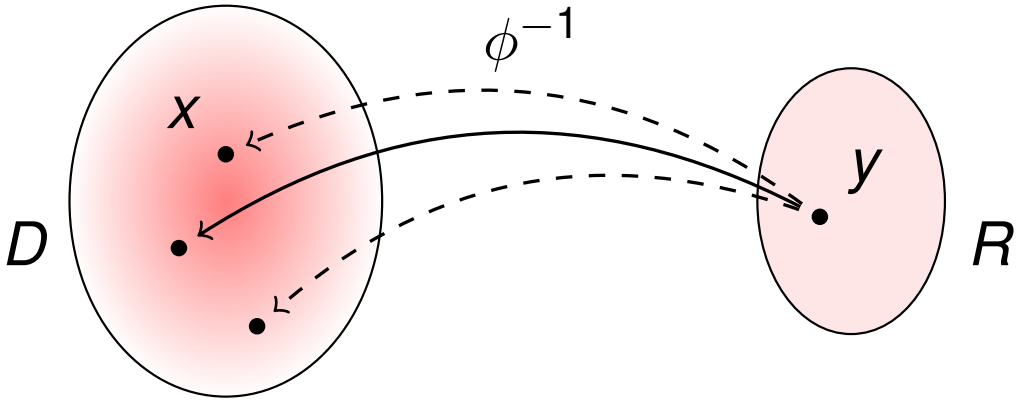

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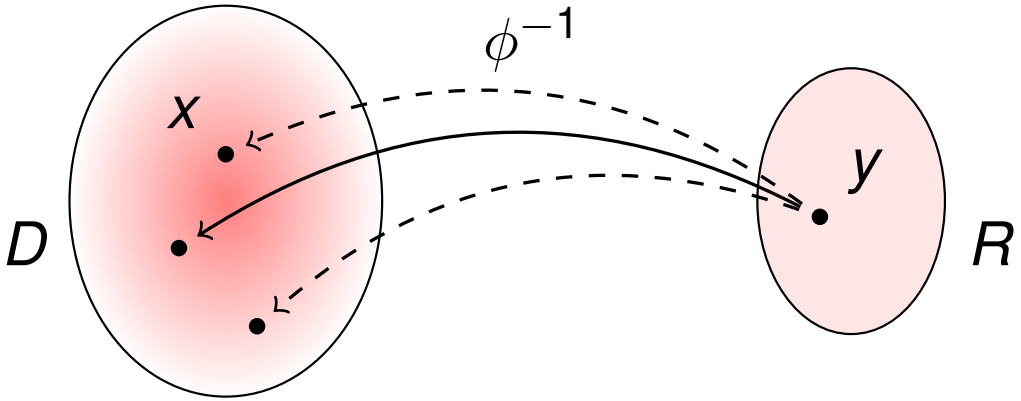

- "Hash and sign:" $pk = \phi$, $sk = \phi^{-1}$, hash $H\colon \{0,1\}^* \to R$

$$\mathrm{Sign}(m) := \phi^{-1}(H(m)).$$

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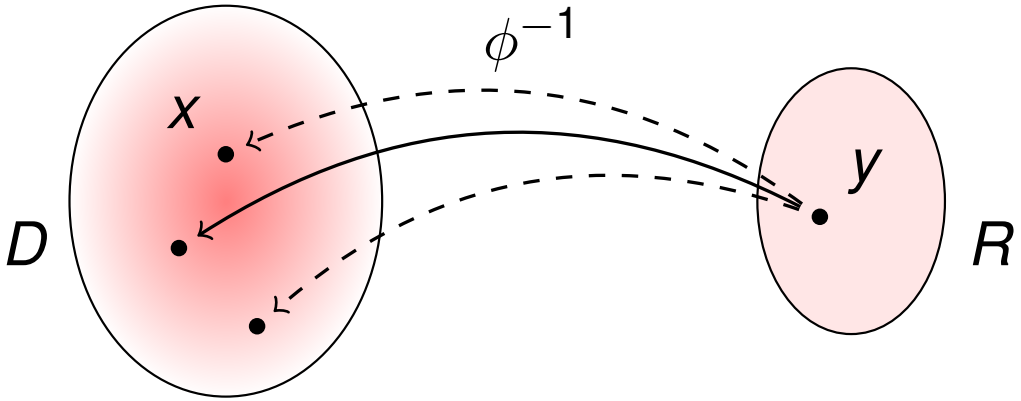

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- Security (ROM): can generate (x, y) in 2 equivalent ways:

REALITY

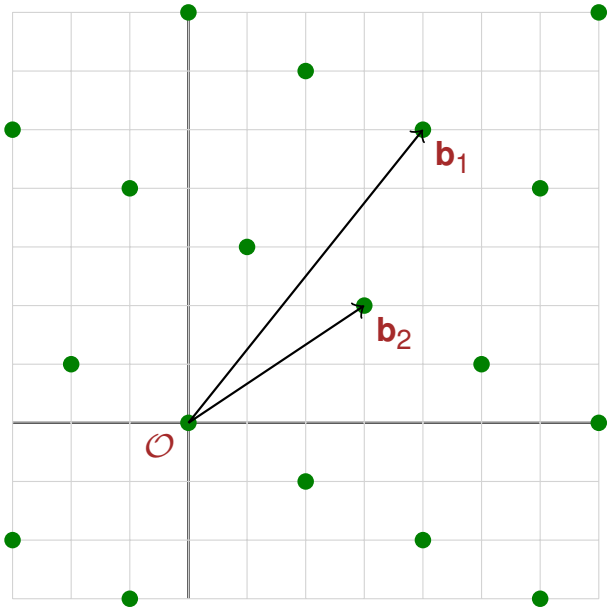
$$x \xleftarrow{\phi^{-1}} y \longleftarrow R$$

PROOF

$$D \longrightarrow x \xrightarrow{\phi} y$$

GPV Signatures, concretely

- $\Lambda \subset \mathbb{Z}^n$ a lattice (full-rank additive subgroup), defined by basis.



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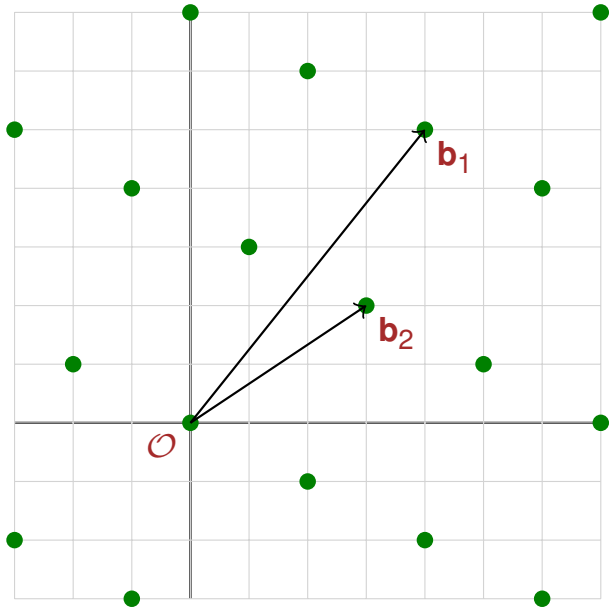

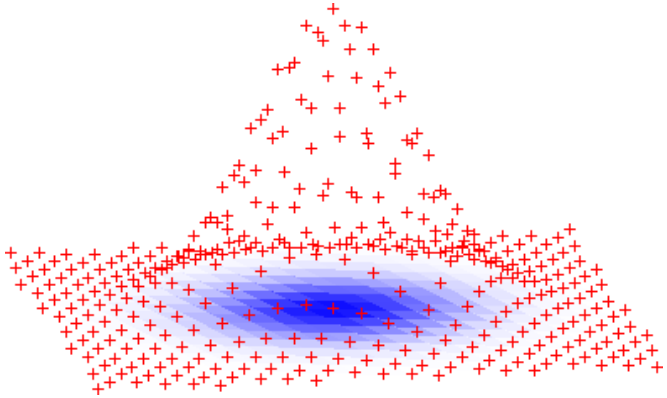

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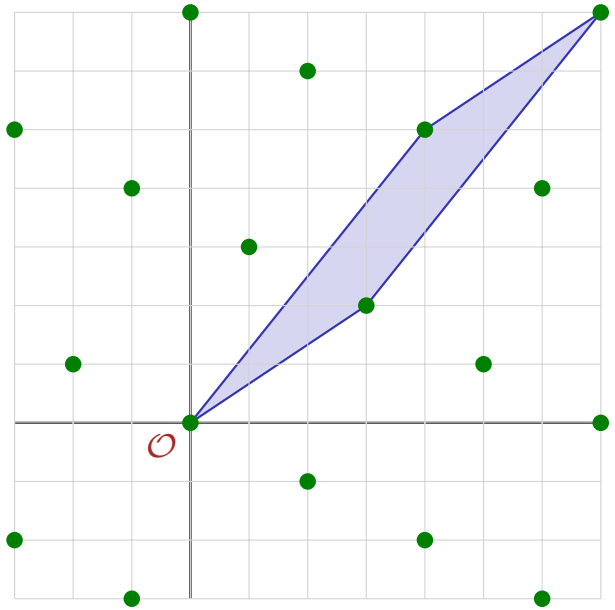

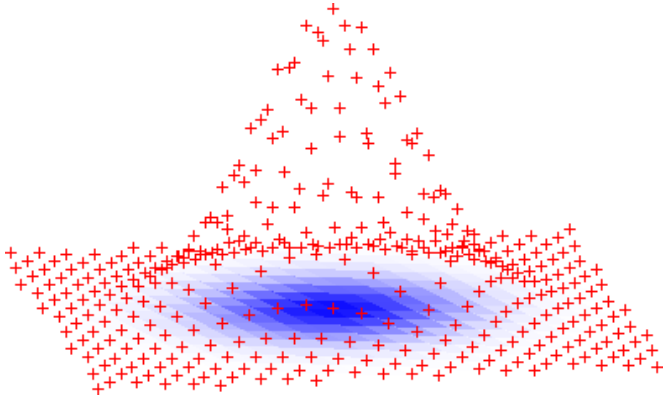

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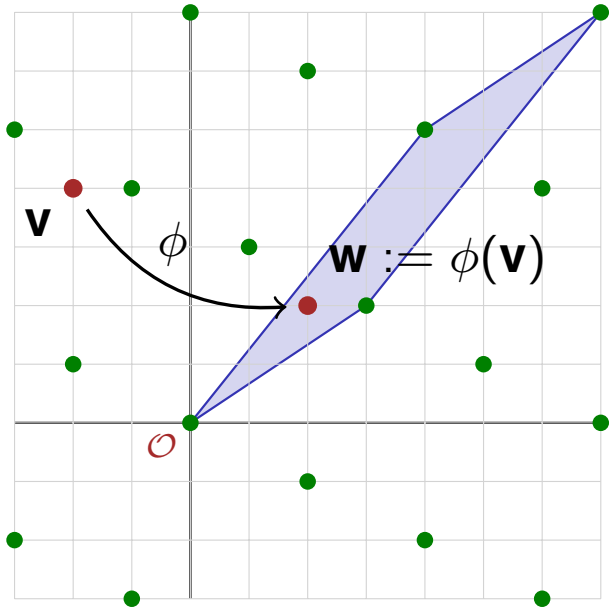

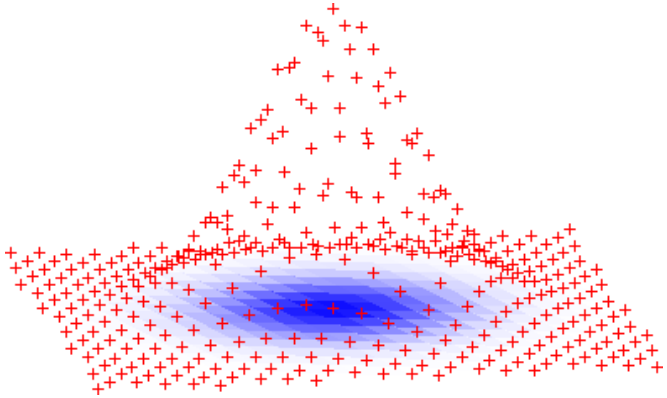

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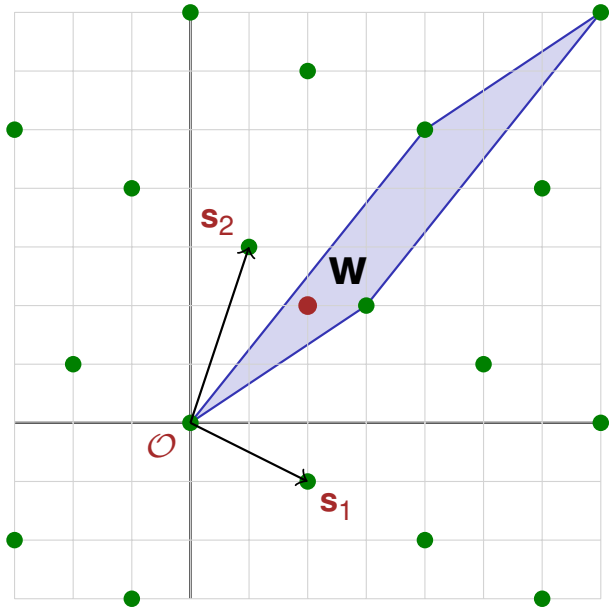

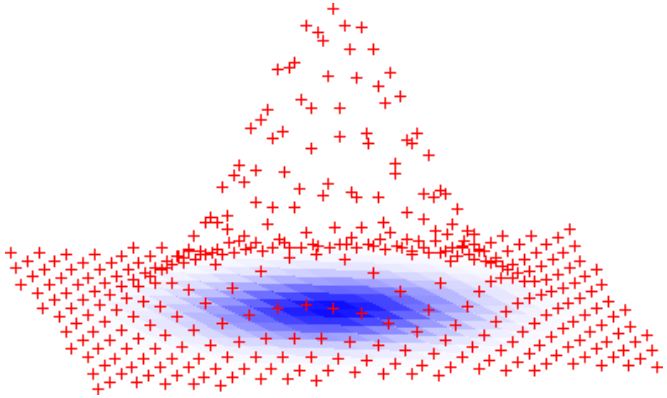

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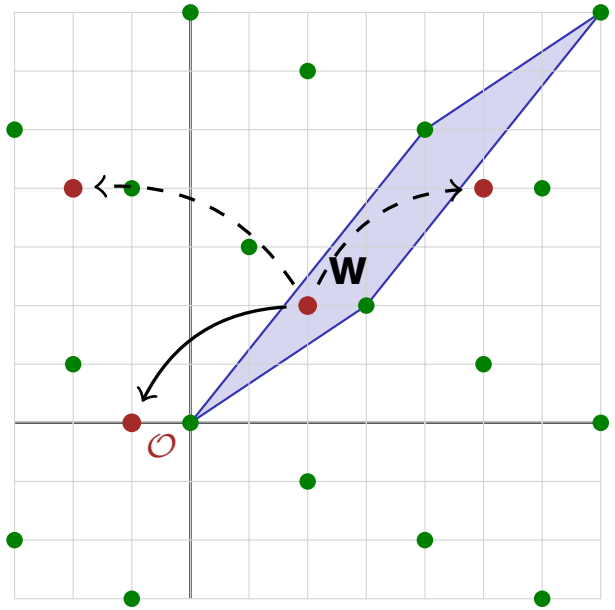

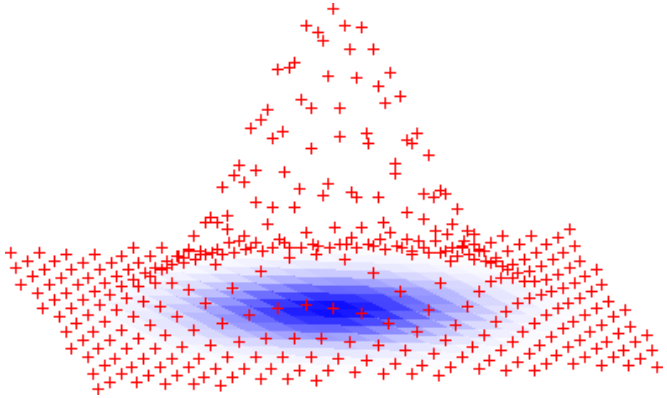

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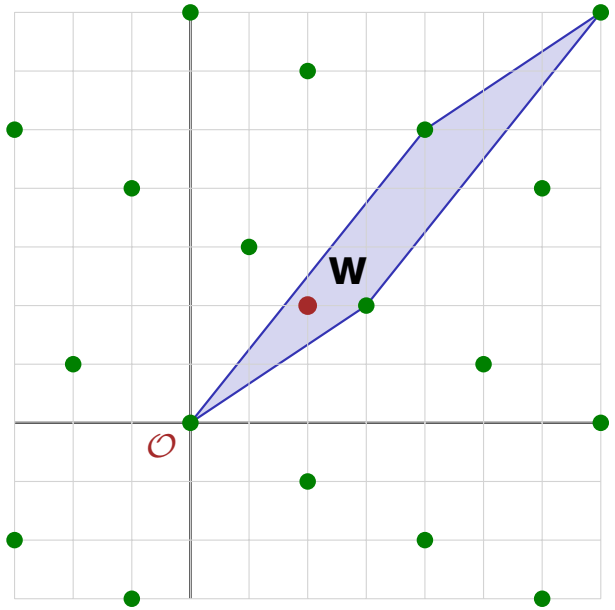
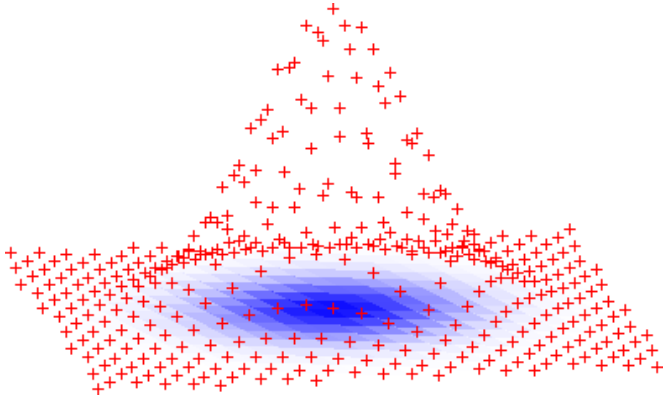
- Sampling from $\Lambda + \mathbf{w}$ without short basis is hard. (How hard depends on Gaussian parameter.)





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GPV sign/verify algorithms: $H\colon \{0,1\} \to \mathbb{Z}^n/\Lambda$

$$\begin{aligned} \text{Sign}(m) &:= \text{short vector in } \Lambda + H(m) \\ \text{Verify}(\sigma) &:= 1 \quad \text{iff} \quad \sigma \bmod \Lambda = H(m) \end{aligned}$$

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- For $a, b \in \mathbb{Z}$, define signature on $am_1 + bm_2$ to be
  $$\sigma := a\sigma_1 + b\sigma_2.$$
  $\Rightarrow \sigma$ is short (if a, b small), $\sigma \bmod \Lambda = am_1 + bm_2$.

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- Malicious DB outputs $18.0 = s_1 - s_2$ and signature $\sigma = \sigma_1 - \sigma_2$.
- σ authenticates 18, but 18 is not the mean!

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Use a second lattice to authenticate functions:

- $\Lambda_2 \subset \mathbb{Z}^n$ distinct from $\Lambda_1 := \Lambda$.
  - require $\Lambda_1 + \Lambda_2 = \mathbb{Z}^n$
- Map $\phi_2 \colon \mathbb{Z}^n \to \mathbb{Z}^n / \Lambda_2$ given by $\phi_2(\mathbf{v}) := \mathbf{v} \bmod \Lambda_2$.

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Encode functions f as elements $\langle f \rangle \in \mathbb{Z}^n/\Lambda_2$.
Sign functions by computing

$$\mathrm{Sign}(f) := \text{short vector in } \Lambda_2 + \langle f \rangle.$$

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- $\Lambda_2 \subset \mathbb{Z}^n$ distinct from $\Lambda_1 := \Lambda$.
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- Map $\phi_2 \colon \mathbb{Z}^n \to \mathbb{Z}^n/\Lambda_2$ given by $\phi_2(\mathbf{v}) := \mathbf{v} \bmod \Lambda_2$.

Encode functions f as elements $\langle f \rangle \in \mathbb{Z}^n/\Lambda_2$.
Sign functions by computing

$$\mathrm{Sign}(f) := \text{short vector in } \Lambda_2 + \langle f \rangle.$$

If encoding $\langle \cdot \rangle$ is linear, (i.e., $\langle f \rangle + \langle g \rangle = \langle f + g \rangle$)
then signature is a linear operator on the space of functions.

How do we encode functions?

$k :=$ number of messages input to a function.

$$\left\{ \begin{array}{c} \text{Linear functions in} \\ k \text{ variables} \end{array} \right\} \quad \text{generated by} \quad \left\{ \begin{array}{c} k \text{ projection} \\ \text{functions } x_i \end{array} \right\}$$

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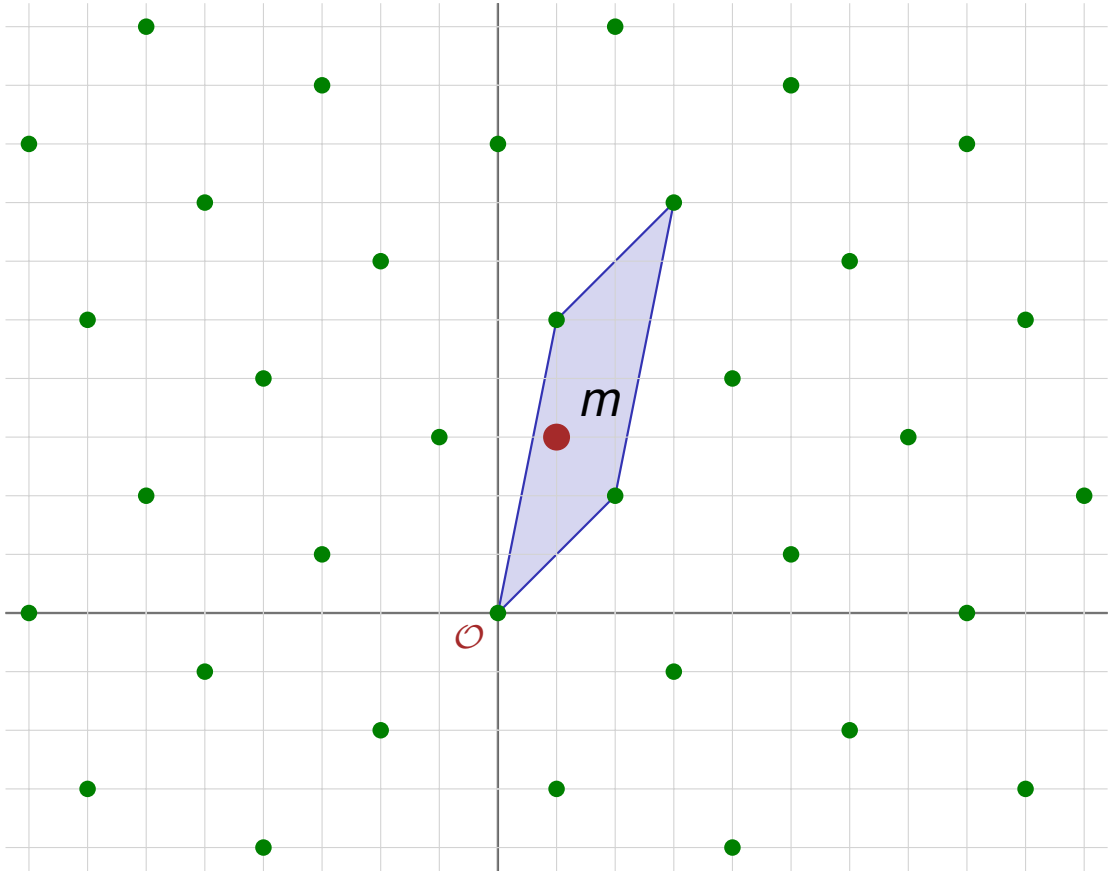
  - Encoding is one-way.
  - Sweeping under the rug (for now): where do the c_i live? (think of c_i as small integers)

“Intersection method” binds messages to functions

- Messages $m \in \mathbb{Z}^n/\Lambda_1$.



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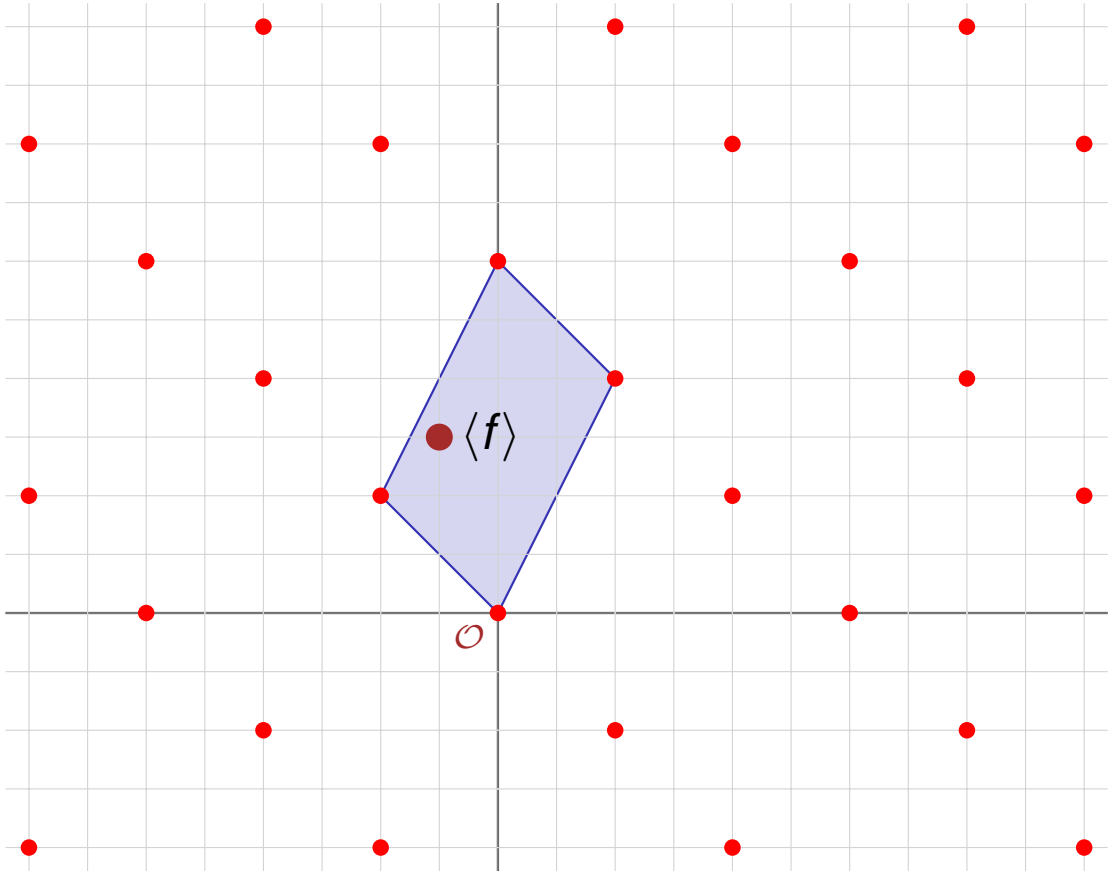

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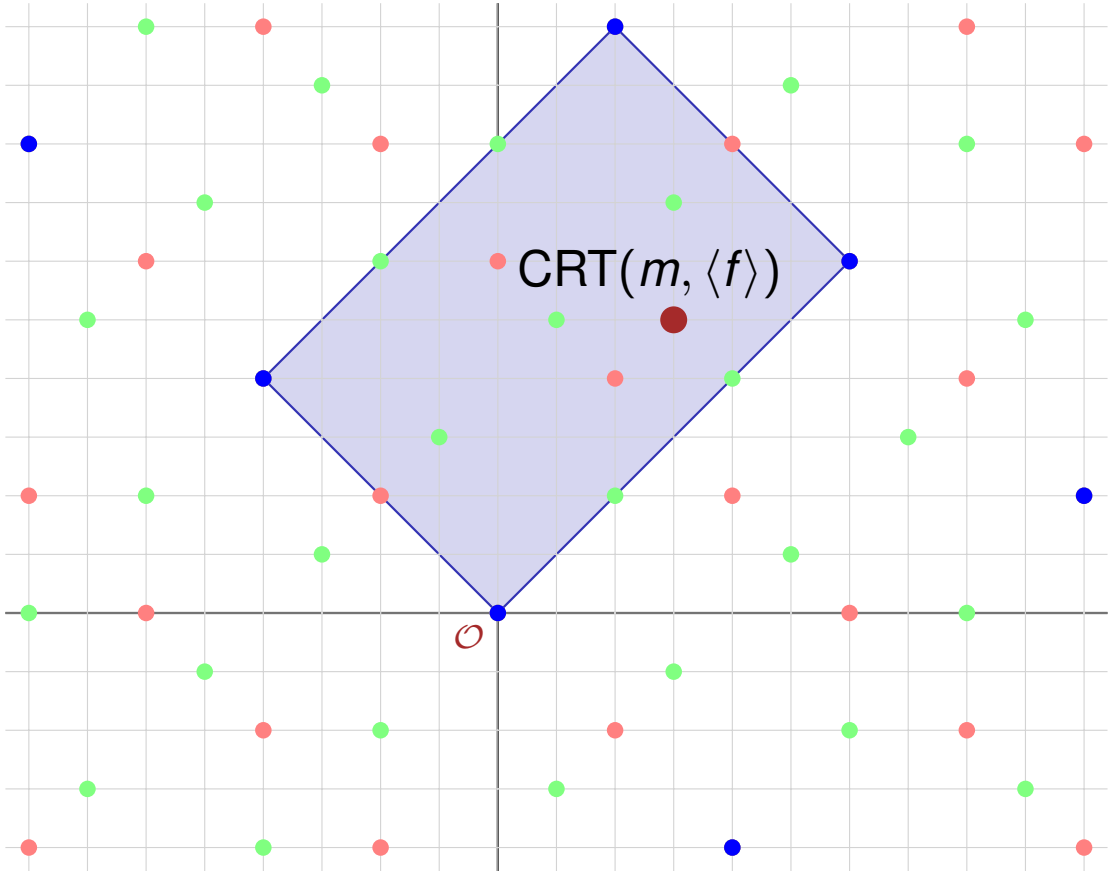

$\mathrm{CRT}(m, \langle f \rangle)$
$= m \bmod \Lambda_1$
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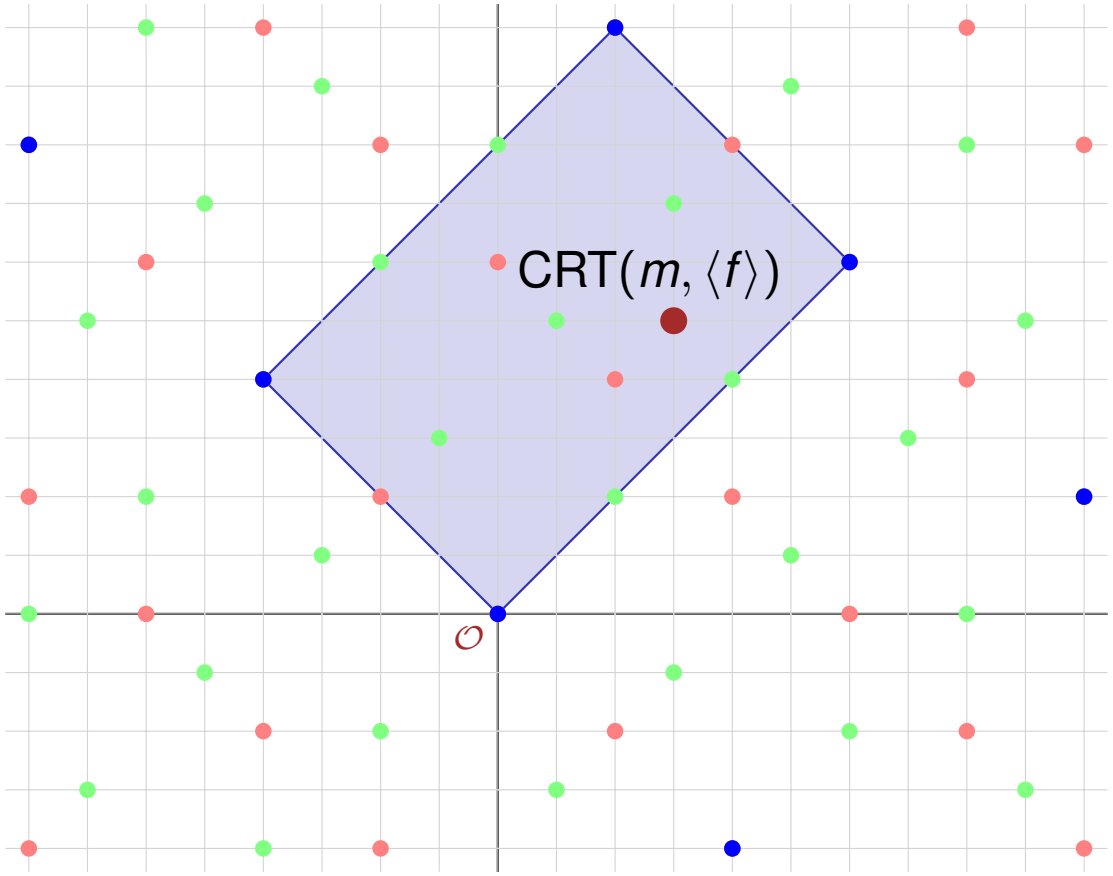

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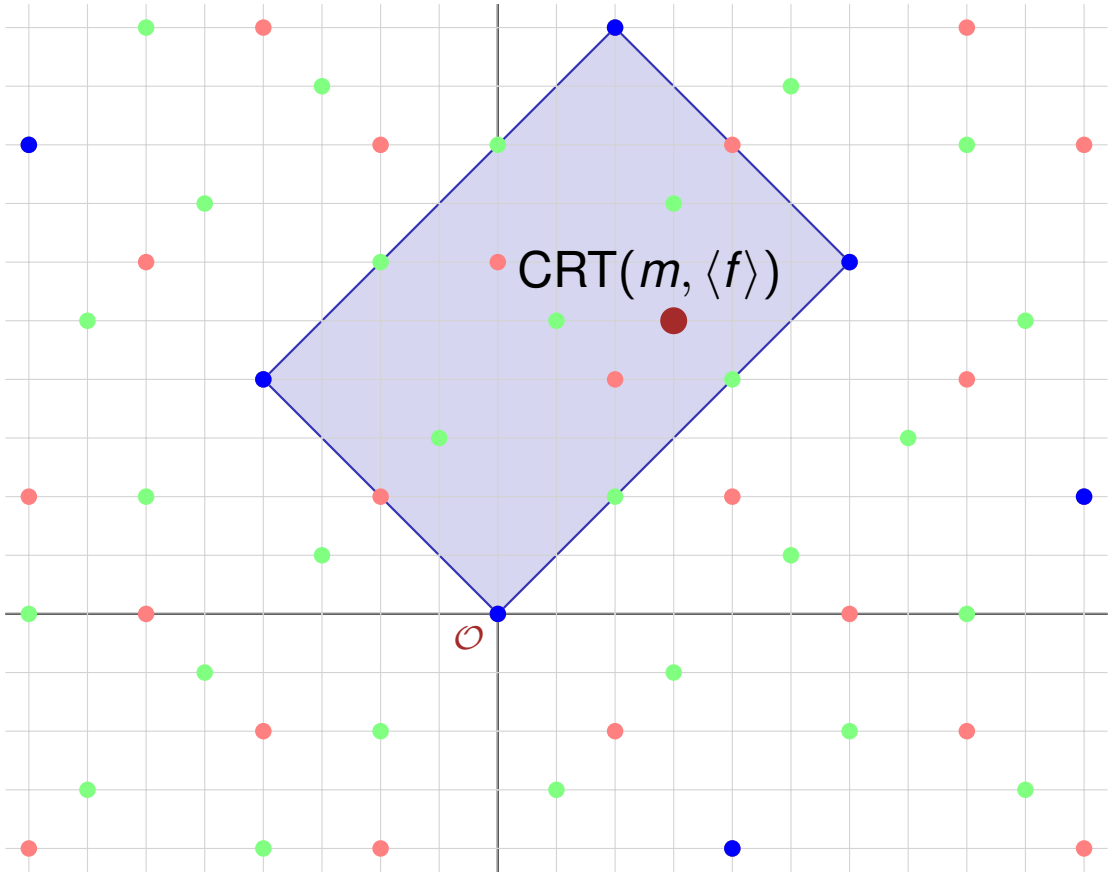

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$$\begin{aligned} \text{Sign}(m) &:= \text{short vector in } (\Lambda_1 \cap \Lambda_2) + \text{CRT}(m, \langle f \rangle) \\ \text{Verify}(\sigma) &:= 1 \quad \text{iff} \quad (\sigma \bmod \Lambda_1 = m) \text{ and } (\sigma \bmod \Lambda_2 = \langle f \rangle) \end{aligned}$$

Linearly homomorphic signature scheme

- KeyGen(n):
  - $pk =$ Lattices $\Lambda_1, \Lambda_2 \subset \mathbb{Z}^n$, Gaussian parameter β
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  1. σ mod $\Lambda_1 = m$,
  2. σ mod $\Lambda_2 = \langle f \rangle = \sum c_i \alpha_i$,
  3. σ sufficiently short: $\|\sigma\| < \beta\sqrt{n} \cdot k \cdot \max\{|\mathrm{lift}(c_i)|\}$.

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Same functionality as *network coding* signatures [BFKW09,GKKR10], except p can be small (even $p = 2$).

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Adversary wins if f admissible, σ^* verifies for (τ^*, m^*, f), and

1. τ^* not obtained in response to a query, or
2. $\tau^* = \tau$ for query $(m_1, \ldots, m_k)$, and $m^* \neq f(m_1, \ldots, m_k)$.

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Proof outline

1. Answer signature queries without a basis of $\Lambda_1 \cap \Lambda_2$.
2. Use adversary's forgery to produce a short vector in Λ_2.

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- Simulation is indistinguishable from real system.

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Conclusion: $\sigma^* - \sigma$ is (1) nonzero, (2) in Λ_2, (3) short.

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Proof outline

2. Use forgery to produce a short nonzero vector in Λ_2.

- Adversary outputs forgery $(\tau^*, m^*, \sigma^*, f = \sum c_i x_i)$.
- Suppose $\tau^*, \sigma_1, \ldots, \sigma_k$ answer query $m_1, \ldots, m_k$.
- Compute $\sigma := \sum c_i \sigma_i =$ "real" sig on $f(m_1, \ldots, m_k)$.
- Validity of forged σ^* and authenticity of "real" σ means:
  1. $\sigma^* \bmod \Lambda_1 = m^* \neq f(m_1, \ldots, m_k) = \sigma \bmod \Lambda_1$
  2. $\sigma^* \bmod \Lambda_2 = \sum c_i \alpha_i = \sigma \bmod \Lambda_2$
  3. σ^* is short and σ is short

Conclusion: $\sigma^* - \sigma$ is (1) nonzero, (2) in Λ_2, (3) short.

- If τ^* not obtained from a query, sign random messages m_i and perform same analysis.

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Privacy property: derived signature on $f(m_1, \ldots, m_k)$ reveals nothing about $m_1, \ldots, m_k$ beyond value of f.

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Specifically: given data sets

$$\vec{m} = (m_1, \ldots, m_k), \qquad \vec{m}' = (m'_1, \ldots, m'_k)$$

and admissible function f with

$$f(\vec{m}) = f(\vec{m}'),$$

even unbounded adversary cannot distinguish derived signature on $f(\vec{m})$ from derived signature on $f(\vec{m}')$.

Privacy theorem

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Linearly homomorphic signatures are private.

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Proof idea

- Distribution of derived signature on $f(\vec{m})$ depends only on f and $f(\vec{m})$, not on $\vec{m}$.
- If $f(\vec{m}) = f(\vec{m}')$, distributions of derived sigs are identical.

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Key fact in proof: distribution of linear combination of Gaussian samples is also Gaussian.

- Sigs on m_i sampled from discrete Gaussian distribution.

Polynomially Homomorphic Signatures from Ideal Lattices

Extending the system

Linearly homomorphic scheme: messages in $\mathbb{Z}^n/\Lambda_1$, signatures in $\mathbb{Z}^n$.

$\phi\colon \mathbb{Z}^n \to \mathbb{Z}^n/\Lambda_1$ given by $\mathbf{v} \mapsto (\mathbf{v} \bmod \Lambda_1)$ is a linear map, so we can add both before and after applying ϕ.

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New idea: what if $\mathbb{Z}^n$ has a ring structure and Λ_1 is an ideal?

Then ϕ is a ring homomorphism, so we can add or multiply both before and after applying ϕ.

Setup for polynomial system [G09]

Fix monic, irreducible $F(x) \in \mathbb{Z}[x]$ of degree n.

- $R := \mathbb{Z}[x]/(F(x))$ gives a ring structure on $\mathbb{Z}^n$:

$$\begin{array}{rcl} \text{(coordinates of vectors)} & \leftrightarrow & \text{(coefficients of polynomials mod } F\text{)} \\ (a_0, \ldots, a_{n-1}) & \leftrightarrow & a_0 + a_1 x + \cdots + a_{n-1} x^{n-1} \end{array}$$

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- Encoding uses lift of f to $\hat{f} \in \mathbb{Z}[x_1, \ldots, x_k]$:
  f encoded as $\hat{f}(\alpha_1, \ldots, \alpha_k)$ mod q.

The polynomial system, concretely

- KeyGen(n):
  - $F(x) \in \mathbb{Z}[x]$ degree n
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  - $pk =$ prime ideals $\mathfrak{p}, \mathfrak{q} \subset R$, Gaussian parameter β
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Products of short elements

Why is a product of short elements of R short?

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[G09,G10]: parameter γ_F measures how much multiplication in R increases length:

$$\gamma_F := \sup_{u,v \in R} \frac{\|u \cdot v\|}{\|u\| \cdot \|v\|}.$$

- Product of d elements of length $< \beta$ has length $< \gamma_F^{d-1} \beta^d$.
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Lots of $F(x)$ have small γ_F:

- Cyclotomic polynomials $\Phi_\ell(x)$, ℓ prime or $\ell = 2^a 3^b$.
- $F(x) = x^n - h(x)$, $\deg(h) < n/2$.
- Others...

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- $\{u \cdot w \cdot x^i\}_{i=0}^{n-1}$ is a short basis of $\mathfrak{p} \cdot \mathfrak{q}$.
- $\mathfrak{p}, \mathfrak{q}$ are principal prime ideals — not uniform.
  - Most ideals are not principal.

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- If $\beta, \gamma_F, k, y \in \text{poly}(n)$ and $d = O(1)$, then derived signature length is poly(n). (p is exponential in n)
- For fixed n, bit length of derived signatures is linear in d, logarithmic in k.

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- $\mathfrak{q}$ is a principal prime ideal.
  - Producing a short generator of arbitrary $\mathfrak{q}$ is a classical problem in algorithmic number theory.
- Distribution of $\mathfrak{q}$ not well-studied.
  - Want $\mathfrak{q}$ in distribution that admits a worst-case reduction.

Open questions

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   (i.e., leak no information about original messages)
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Thank you!

Thanks also to Chris Peikert for help with graphics.